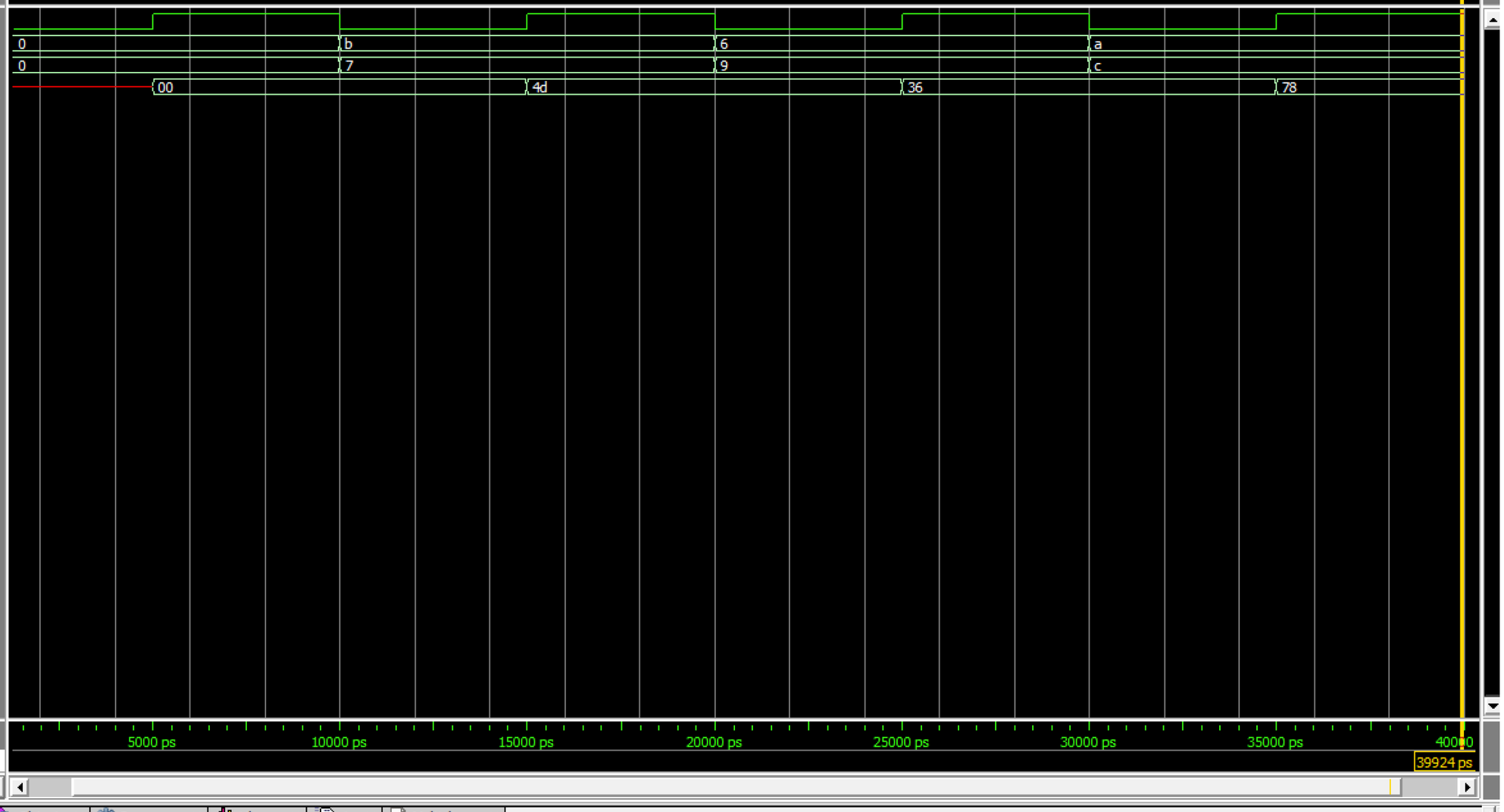
Questa Altera Starter FPGA Edition-64 2025.2

File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help

Help | 100 ps | Layout Simulate | ColumnLayout Default | Search:

Wave - Default

	Msgs
/Multiplier_tb/clk	1'h1
/Multiplier_tb/M	4'ha
/Multiplier_tb/Q	4'hc
/Multiplier_tb/R	8'h78
Now	40000 ps
Cursor 1	39924 ps



Transcript | Wave | Dataflow | Objects | Processes | Library | sim | Multiplier.v

Example of how the multiplier works:

For b×7 [1011×0111],

M=1011, Q=0111

i=0

C	A	q
0	0000	0111

q[1]=1 -> A=M+A

C	A	q
0	1011	0111

Shift C, A and q right by 1 bit:

C	A	q
0	0101	1011

M=1011, Q=0111

i=1

C	A	q
0	0101	1011

q[1]=1 -> A=M+A

C	A	q
1	0000	1011

Shift C, A and q right by 1 bit:

C	A	q
0	1000	0101

M=1011, Q=0111

i=2

C	A	q
0	1000	0101

q[1]=1 -> A=M+A

C	A	q
1	0011	0101

Shift C, A and q right by 1 bit:

C	A	q
0	1001	1010

M=1011, Q=0111

i=3

C	A	q
0	1001	1010

q[0]=1 -> No addition

C	A	q
0	1001	1010

Shift C, A and q right by 1 bit:

C	A	q
0	0100	1101

Final result: R = 0100 1101 = 77

A q

b(11) × 7 = 77